

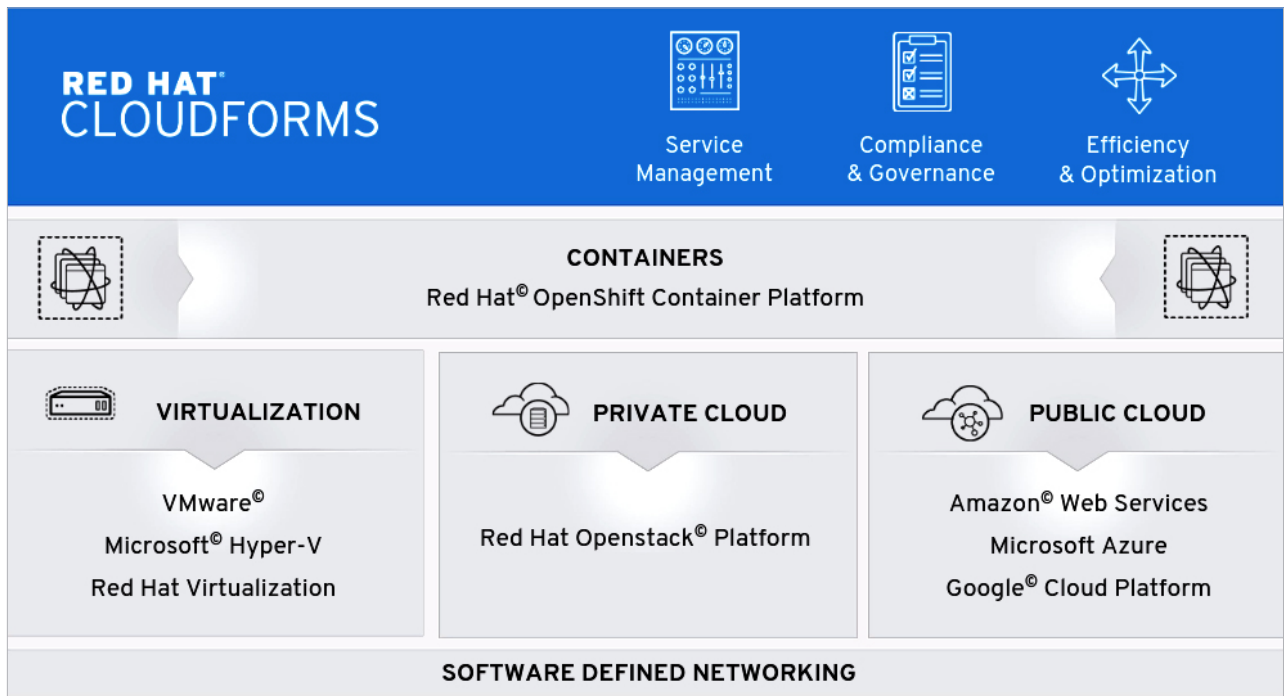


Figure 6: Red Hat CloudForms (Image source: Red Hat documentation)

The core features of ManageIQ include the following.

Continuous discovery: Manages the inventory of cloud services including virtualisation, containers, network, storage, and compute services. It automatically updates when there is a change in the environment and serves as a ready-to-use configuration management database (CMDB) for the assessment of cloud readiness such as BMC Remedy or CAST.

Service catalogue: It manages the full life cycle of a cloud service, which includes policy management, compliance checks, delegated operations, and chargeback for cost management reports. End users can order services for provisioning from the given service catalogue.

Auto-discovery and compliance: Scans through the contents of cloud services using SmartState analysis, including VMs and containers, and creates security and compliance policies.

Cloud optimisation: Collects business and technical telemetry, and

helps to prepare visualisation reports for cloud service utilisation and chargeback summary. This helps to build a capacity-planning model for the enterprise.

Advantages of using cloud management tools

Cloud management services have the following five cross-functional attributes:

- AI to provide advanced optimisation facilities using ML based solutions for analytics and anomaly detection.
- Brokerage to enable the relation between cloud provider, cloud services and consumers for seamless cloud migration.
- Automation to enable facilities like infrastructure-as-code (IaC), automatic code review, and package and delivery of cloud services.
- Governance to provide the mechanism for user management and role definition.
- Life cycle to enforce the governing principles in cloud adoption in a systematic way with best standards.

Infrastructure-as-code helps multi-cloud management since preparing provisioning scripts for each cloud is inconsistent and inefficient. Though cloud native facilities like CloudFormation in AWS, ARM templates in Azure and CDM templates in GCP help to achieve this infrastructure-as-code, they are mostly native to the cloud platform and don't have the agility and flexibility for multi-cloud management. However, Google Anthos, HashiCorp Terraform and Red Hat Ansible templates can help with multi-cloud management and with developing zero-touch deployment scripts and provisioning templates for cloud tower setup. All three tools lead equally in this space for creating, changing and provisioning infrastructure for any cloud platform, including service management, security, compliance and optimisation. Red Hat CloudForms has five integral components as follows.